OS Unit 4 Orange Problem

Name: Sathvik Karthik Malali
SRN: PES1UG24CS616
Semester: 4
Section: K

Phase 1

Screenshot 1A

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Screenshot 1B

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Phase 2

Screenshot 2A

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Screenshot 2B

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -n 20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Phase 3

Screenshot 3A

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes init
Initialized empty PES repository in .pes/
→  PES1UG24CS616-pes-vcs git:(main) ✗ echo "hello" > file1.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ echo "world" > file2.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes add file1.txt file2.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  README.md
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
```

Screenshot 3B

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776325356 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776325363 6 file2.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Phase 4

Screenshot 4A

PES1UG24CS616-pes-vcs — zsh

```
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes init
Initialized empty PES repository in .pes/
→  PES1UG24CS616-pes-vcs git:(main) ✗ echo "Hello" > hello.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes add hello.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes commit -m "Initial commit"
[5dcd5788a6bd67df2a61447c733d0833d57db1bc7a3bd8d6a0af8e76c6ccfcfb] Initial commit
Committed: 5dcd5788a6bd... Initial commit
→  PES1UG24CS616-pes-vcs git:(main) ✗ echo "World" >> hello.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes add hello.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes commit -m "Add world"
[2273db7cbb011d8a7542872440bbf53da5b488815de27bbe4d0f88de82bb7da2] Add world
Committed: 2273db7cbb01... Add world
→  PES1UG24CS616-pes-vcs git:(main) ✗ echo "Goodbye" > bye.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes add bye.txt
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes commit -m "Add farewell"
[b3bf4ebe118aee67a6e96b3f8fd05276353b82c045e83e171d1a1d1a43f4ffe4] Add farewell
Committed: b3bf4ebe118a... Add farewell
→  PES1UG24CS616-pes-vcs git:(main) ✗ ./pes log
commit b3bf4ebe118aee67a6e96b3f8fd05276353b82c045e83e171d1a1d1a43f4ffe4
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325587

    Add farewell

commit 2273db7cbb011d8a7542872440bbf53da5b488815de27bbe4d0f88de82bb7da2
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325578

    Add world

commit 5dcd5788a6bd67df2a61447c733d0833d57db1bc7a3bd8d6a0af8e76c6ccfcfb
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325551

    Initial commit

→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Screenshot 4B

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/22/73db7cbb011d8a7542872440bbf53da5b488815de27bbe4d0f88de82bb7da2
.pes/objects/5d/cd5788a6bd67df2a61447c733d0833d57db1bc7a3bd8d6a0af8e76c6ccfcfb
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/b3/bf4ebe118aee67a6e96b3f8fd05276353b82c045e83e171d1a1d1a43f4ffe4
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Screenshot 4C

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ cat .pes/refs/heads/main
b3bf4ebe118aee67a6e96b3f8fd05276353b82c045e83e171d1a1d1a43f4ffe4
→  PES1UG24CS616-pes-vcs git:(main) ✗ cat .pes/HEAD
ref: refs/heads/main
→  PES1UG24CS616-pes-vcs git:(main) ✗
```

Final Screenshots (test-integration)

```
PES1UG24CS616-pes-vcs — zsh
→  PES1UG24CS616-pes-vcs git:(main) ✗ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
[54935db1becaa3782a419d54df62073ee80b3b6b572eed7b12e8611067c3118c] Initial commit
Committed: 54935db1beca... Initial commit

Log after first commit:
commit 54935db1becaa3782a419d54df62073ee80b3b6b572eed7b12e8611067c3118c
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325905

    Initial commit


--- Second Commit ---
[bd7a947ea450ce64405dbb858d17cf8bad3ae06ea2b8bcf000c9c61180dfda14] Update file.txt
Committed: bd7a947ea450... Update file.txt

--- Third Commit ---
[b1399209c1b337262500566135ae8ae31e4ae681a3c828b91cbe28dee7cbfadf] Add farewell
Committed: b1399209c1b3... Add farewell

--- Full History ---
commit b1399209c1b337262500566135ae8ae31e4ae681a3c828b91cbe28dee7cbfadf
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325905

    Add farewell
```

```
commit bd7a947ea450ce64405dbb858d17cf8bad3ae06ea2b8bcf000c9c61180dfda14
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325905

    Update file.txt

commit 54935db1becaa3782a419d54df62073ee80b3b6b572eed7b12e8611067c3118c
Author: Sathvik Malali <PES1UG24CS616>
Date:   1776325905

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
b1399209c1b337262500566135ae8ae31e4ae681a3c828b91cbe28dee7cbfadf

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/54/935db1becaa3782a419d54df62073ee80b3b6b572eed7b12e8611067c3118c
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/399209c1b337262500566135ae8ae31e4ae681a3c828b91cbe28dee7cbfadf
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/bd/7a947ea450ce64405dbb858d17cf8bad3ae06ea2b8bcf000c9c61180dfda14
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
➜  PES1UG24CS616-pes-vcs git:(main) ✗
```

Phase 5

Q5.1

Checkout updates .pes/HEAD to point to the target branch, reads the commit hash from .pes/refs/heads/<branch>, loads its tree, and rewrites the working directory to match that tree. It also updates the index to reflect the new snapshot. The complexity comes from safely handling file overwrites, deletions, and directory structure changes while avoiding loss of uncommitted work.

Q5.2

For each tracked file, compare the working directory state with the index by recomputing the hash. If a file is modified relative to the index and also differs between the current tree and target tree, checkout must refuse. This detects conflicts by comparing working directory vs index and index vs target tree.

Q5.3

In detached HEAD, commits are created normally but no branch points to them, so they become unreachable once you move away. The user can recover them if they know the commit hash by creating a new branch pointing to it, or by checking it out again before it is garbage collected.

Phase 6

Q6.1

Start from all branch heads and traverse commits, trees, and blobs using DFS or BFS, marking all reachable object hashes in a hash set. Then scan the object store and delete any object not in the reachable set. With 100k commits and shared history, traversal is roughly 100k commits plus their trees and blobs, not multiplied by branches.

Q6.2

If GC runs while a commit is being created, it may delete objects (like blobs or trees) that are written but not yet referenced by any commit. The commit then ends up pointing to missing objects. Git avoids this by writing objects before updating refs, delaying deletion of recent objects, and using safeguards like timestamps and locks to prevent such races.